

## SOLUTION REQUIREMENTS

|                             |                                                                                                        |                                   |
|-----------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Date:</b> 18th July 2026 | <b>Project Name:</b> Book a Doctor App                                                                 | <b>Team Lead:</b> Kilari Ushodaya |
| <b>Domain:</b> Healthcare   | <b>Team Members:</b> Shyam Prasad Mantri, Madhu Hasitha Kondeti, Nanda Kumar Madugula, Mantena Sirisha |                                   |

### Functional Requirements (FR)

| Req ID | Module Name    | Functional Requirement Description                           | Priority |
|--------|----------------|--------------------------------------------------------------|----------|
| FR-01  | User Auth      | Multi-role authentication for Patients, Doctors, and Admins. | High     |
| FR-02  | Doctor Catalog | Search doctors by name, specialty, location, and rating.     | High     |
| FR-03  | Symptom Triage | Symptom query parsing to recommend medical specialists.      | High     |
| FR-04  | Slot Booking   | Real-time calendar slot picking with instant slot locking.   | High     |
| FR-05  | Document Vault | Encrypted PDF/image upload vault for diagnostic reports.     | High     |

### Non-Functional Requirements (NFR)

| Category    | Target Requirement Metric | Target Specification                                  |
|-------------|---------------------------|-------------------------------------------------------|
| Performance | API Response Latency      | Average response < 300 ms; triage < 1,000 ms.         |
| Security    | Data Encryption & Storage | AES-256 encryption at rest; TLS 1.3 HTTPS in transit. |
| Scalability | Concurrent User Load      | Supports up to 5,000 active concurrent users.         |
| Compliance  | Privacy Regulations       | Strict HIPAA and GDPR healthcare data compliance.     |